

# AMOGH SOMISETTY

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EE junior seeking a Fall 2026 GPU / ML-hardware internship or co-op: GPU power and thermal characterization (blind-validated on 72 production H100s), carrier-board design for Jetson AGX Orin edge AI, and RTL-level RISC-V design.

## EDUCATION

**California Polytechnic State University, San Luis Obispo**

B.S. Electrical Engineering

Expected June 2028

San Luis Obispo, CA

- **Coursework:** Computer Architecture, Digital Logic Design, Embedded Systems, Circuits I–III, Signals & Systems

## EXPERIENCE

**Founder & Lead Researcher**

Theta — open-source GPU thermal-power forensics (ThermalOS research)

May 2026 – Present

San Luis Obispo, CA

- Built and shipped **Theta**, an open-source GPU reliability agent (`pip install runtheta`) computing peer-relative thermal resistance ( $R_\theta = \Delta T/P$ ) from NVML/DCGM telemetry to separate cooling degradation from workload effects on live clusters.
- Blind-flagged 3 degraded units across **72 H100 SXM5 GPUs** (Princeton Della cluster) via two-way median-polish peer z-scores: top anomaly  $z = +15.6$  (80 °C at 653 W vs. 57–66 °C node-mates), two invisible to temperature thresholds; detection replay hit at **6 minutes** vs. a multi-day ticket loop.
- Predicted a degrading unit blind: 3 months later, Princeton’s own DCGM diagnostic independently re-confirmed the flag at **+47.8%** thermal resistance and **–2.6%** GFLOPS at matched power, caught pre-throttle.
- Characterized  $R_\theta$  across four GPU generations (T4, V100, A100, H100) on real telemetry; controlled thermal-memory experiments (8,700+ samples) showed a 2 °C ambient delta drives a **3.5×** change in power-recovery time at <2% within-group CV.
- Shipped a 6-layer detection stack (classifier ensemble, predictive alerting, fleet correlator, ECC/micro-throttle monitoring, Isolation Forest drift critic, fault-curve diagnosis of 6 failure modes); preparing a publication targeting ICPE 2027, advised by ex-Intel silicon-validation faculty.

**Edge-AI Hardware Designer — Carrier Boards & Sensor Modules**

Poly UAS (Cal Poly Unmanned Aerial Systems)

Apr 2026 – Present

San Luis Obispo, CA

- Own carrier-board and sensor-module hardware design for the team’s NVIDIA Jetson AGX Orin edge-AI stack, moving the aircraft from ground-controlled flight to onboard real-time computer vision and autonomous decision-making.
- Design schematics and PCB layout for the Orin carrier board and supporting sensor modules: power delivery, sensor interfacing, and size/weight/power constraints for flight hardware.

**Lab Operations Intern: Systems, Automation & Validation**

Natera

June 2025 – Sept 2025

Pleasanton, CA

- Cut validation and packaging cycle time from ~20 min to 10–12 min (>40%) by mapping the workflow and eliminating redundant verification steps in a high-throughput production line.
- Built Python automation for barcode validation and pathology-report retrieval, removing manual lookups across the team.
- Ran root-cause and failure analysis on integrated electrical, mechanical, and software systems; authored reference documentation adopted by the team to prevent recurrence.

## TECHNICAL PROJECTS

**RISC-V CPU (RV32I) on FPGA** | SystemVerilog, Vivado, Basys-3

- Designed a multi-cycle RV32I CPU from the RTL up (full ISA, 10-operation ALU, FSM control unit, 32×32 register file); synthesized and verified on a Xilinx Basys-3, debugging branch-comparison and immediate-generation logic through waveform analysis and iterative timing fixes.

**Audio ADC → DAC Signal Chain on Custom PCB** | AVR, LTspice, Autodesk Fusion

- Designed, simulated, and fabricated a binary-weighted 4-bit DAC on a 2-layer PCB driven by register-level AVR firmware (direct port writes); achieved 62.5 mV/step resolution within ±1 LSB across all 16 codes and caught a solder bridge via continuity check before power-on.

## TECHNICAL SKILLS

**Languages & Tools:** Python, C/C++, SystemVerilog/Verilog, TypeScript, Bash, MATLAB, Git

**GPU & Telemetry:** NVML (pynvml), DCGM, nvidia-smi, Prometheus, SLURM, Linux, GPU power/thermal characterization, anomaly detection

**ML & Data:** scikit-learn, pandas, NumPy, ensemble classifiers, Isolation Forest, survival analysis (lifelines), FastAPI

**Digital & FPGA:** RTL design, FSMs, Vivado synthesis, static timing, waveform debugging

**Embedded & Hardware:** NVIDIA Jetson (AGX Orin), STM32, Arduino/AVR, I2C/SPI/UART, register-level I/O, PCB design (Autodesk Fusion), LTspice, op-amp/analog conditioning, oscilloscope/DMM, SMD/reflow soldering

## LEADERSHIP

**President, Poly Engineering Consulting:** lead client-facing engineering project teams

Sept 2024 – Present

**Lab Assistant, Cal Poly EPIC:** teach embedded systems to precollegiate students

Summer 2026